

1. Introduction

Welcome to your Summative Assessment 2 for Machine Learning in Master of Science in Business Analytics Program. The aim of this assessment is to test your development of awareness and competencies on Machine Learning acquired during the entire course. This assessment is in a report format that will enable you to enhance awareness of Business Analytics approaches, and applications in today's business landscape.

In performing your solution, carefully read each question that is presented in the final part of this task. Make sure you comprehend every item. Refer each question to one or more specific topics presented in the course, in your opinion, relevant to responding. Provide your reply using the particular concepts and knowledge acquired on every topic that best addresses each question.

2. Course Learning Outcomes

#	CLO DESCRIPTION	GLO1	GLO2	GLO3	GLO4	QFE	CLS
CLO1	Critically analyze significant characteristics of data sets to adopt machine learning methods and principles	X				K,S,R	CLS1, CLS2, CLS5, CLS6
CLO2	Critically appraise business problems to select and implement suitable machine learning methods		X			K,S,R	
CLO3	Critically develop advanced problem solving skills to train, test, and validate phases of learning algorithms			X		K,S,R	
CLO4	Formulate solutions to overcome challenges associated with new machine learning problem settings in a business environment				X	K,S,R	

3. Assessment

3.1 Assessment Requirement

- Select a real-world business or societal problem suitable for machine learning analysis.
- Students must obtain instructor approval for the selected topic before beginning the assessment. List your topic in the file at this link: [Topics.docx](#)
- Once finalized, all four questions below must be answered using the same problem and research direction.

Q1

Introduction & Problem Statement (~300 words)

Appraise your selected **business** or societal problem by explaining its relevance, significance, and impact on key stakeholders. Convert the selected problem into a **machine learning** problem using appropriate concepts from the CRISP-DM framework and identify the most appropriate **machine learning** paradigm. Formulate one or more clear research question(s) (RQs) to guide your investigation.

CLO2 (5 Marks)

Q2

Dataset Analysis (~300 words)

Identify a publicly available dataset (e.g., Kaggle, UCI, government open **data**) that supports investigation of the selected problem. **Analyze** the dataset by examining its source, size, structure, key variables, and **data** types. Explain how the selected dataset aligns with the research questions and **machine learning** task. Briefly discuss one or two recent relevant studies similar to this research and identify the datasets used in those studies.

CLO1 (5 Marks)

Q3

Model Development and Comparison (~700 words)

Develop a complete machine **learning** pipeline for your approved research **problem** by applying appropriate **phases** of the CRISP-DM framework. Implement three machine **learning** methods studied in this course using the same pipeline to support fair evaluation; one method is sufficient if the selected task uses unsupervised **learning**. Use at least three appropriate performance metrics or evaluation measures to assess model performance or results, and present them clearly.

Note: Upload your python code on LMS. Submissions without code will not be marked.

CLO3 (25 Marks)

Q4

Discussion (~700 words)

Formulate the key findings of your **machine learning** analysis in relation to the research questions. Elaborate on what the results mean for the selected **business** or societal **problem** and its key stakeholders. Propose actionable recommendations based on the analysis and elaborate on their practical value. Elaborate on the limitations, **challenges**, and ethical considerations of the study. Finally, propose future improvements or directions for continued research.

CLO4 (25 Marks)

Terms of Reference

As a GUIDE ONLY, this section outlines some of the aspects you may consider significant given the brief shown above. Your report should include:

- ADSM cover page
- Table of Contents
- Abstract
- Answers to the assessment requirements above (use suitable headings to identify the response to each question).
- Conclusion
- References: Include a minimum of 10 articles from reputable academic journals or conferences directly relevant to the topics discussed. Each reference



must clearly support the statement or argument where it is cited. Use APA citation style consistently throughout the text. Each citation should include a DOI or a valid weblink. References carry 15% of the total marks.

Complete the tasks fully but concisely and as directly as possible. Follow all specific instructions for individual questions. Answer all tasks in your own words. Do not copy any text from the notes, readings, or other sources. The assignment must be your own work only.

Total Word Count –2000-2500 words

Submission Date: 14 June, 23:59 pm. Turnitin through LMS.

File Name: Please save your file using the following naming convention. Also use the same name in the title field while uploading on the LMS:
MSBA.BUS8405.SP26.FL25.S1.RHAsmt.<Your ID>

Submission Date: June 14, 2026

Rubrics

Rubric: Summative Assessment - Research Report

DESCRIPTOR	A (100.0-93.0)	A- (92.0-88.0)	B+ (87.0-84.0)	B (83.0-80.0)	B- (79.0-77.0)	C+ (76.0-74.0)	C (73.0-72.0)	C- (71.0-70.0)	F (69.0-0.0)
CLO1 Critically analyze significant characteristics of data sets to adopt machine learning methods and principles	Thoroughly identifies and breaks down relevant dataset characteristics (e.g., structure, type, imbalance, noise). Demonstrates insightful understanding of how these characteristics influence method selection. Analysis is well-structured, accurate, and technically precise.	Strong analysis with clear identification of most key characteristics. Explanation of method relevance is mostly accurate and well reasoned.	Good analysis with moderate insight into data characteristics. Method connection is present but could be more robust.	Basic analysis with relevant observations. Connections to method selection are present but lack clarity or depth.	Minimal analytical depth. Characteristics are listed but not well explained or connected to ML choices.	Limited or surface-level analysis. Misinterprets or overlooks important aspects of the dataset.	Very basic and mostly descriptive; lacks meaningful analysis.	Inaccurate analysis. Shows little understanding of dataset relevance.	No meaningful analysis provided; major misinterpretations or entirely missing.
CLO2 Critically appraise business problems to select and implement suitable machine learning methods	Provides a detailed and critical evaluation of the business problem. Selects and justifies the most appropriate ML method(s) based on problem type, data, and objectives. Shows excellent reasoning and alignment between method and application.	Strong evaluation and appropriate method selection. Justification is clear, with only minor gaps in depth	Good appraisal of the problem. Method choice is appropriate with general justification.	Adequate evaluation. Method selection fits but lacks deeper rationale or consideration of alternatives..	Limited use of recent sources, relying on older references (past 10 years). Integration is weak, with formatting issues.	Minimal evaluation. Reasoning for chosen method is vague or unsupported.	Weak or incomplete appraisal of the problem. Poor linkage to method selection.	Incorrect method or flawed justification. Limited understanding of business context.	No valid evaluation or method selection attempted. Off-topic or incorrect throughout.

DESCRIPTOR	A (100.0-93.0)	A- (92.0-88.0)	B+ (87.0-84.0)	B (83.0-80.0)	B- (79.0-77.0)	C+ (76.0-74.0)	C (73.0-72.0)	C- (71.0-70.0)	F (69.0-0.0)
CLO3 Critically develop advanced problem solving skills to train, test, and validate phases of learning algorithms	Designs and implements a complete end-to-end machine learning solution using real-world datasets. Demonstrates advanced problem-solving skills across the full ML pipeline: data understanding, data preprocessing, feature selection, modeling, and evaluation. Justifies the use of multiple ML with coherence between data characteristics, business problem, and algorithm design. Shows deep technical insight and originality using recognized ML platforms producing outcomes that are technically sound and contextually aligned.	Develops and applies a well-structured ML pipeline using real-world data. Covers most major phases, including data understanding, preprocessing, modeling, and evaluation. Uses more than one ML algorithm with reasonable justification, showing alignment with business objectives. Demonstrates strong technical understanding and clear logic using suitable ML platforms with only minor gaps in clarity or originality.	Constructs a working ML model with most key phases addressed (e.g., preprocessing, modeling, evaluation). Applies appropriate algorithm(s) and provides general reasoning for their use. Shows good technical competence with some depth, though integration across the pipeline or business problem could be improved.	Applies a basic ML workflow with essential components present but may lack completeness or depth in certain phases (e.g., minimal feature selection or shallow evaluation). Uses a relevant algorithm with limited justification. Demonstrates acceptable technical understanding but lacks stronger cohesion between business need and solution.	Demonstrates partial implementation of the ML process (e.g., skips preprocessing or model tuning). Method selection is somewhat appropriate but weakly justified. Technical implementation is basic, and application to the business scenario is general or surface-level.	Attempts an ML solution with only limited workflow coverage. Chooses an algorithm but with vague or incomplete rationale. Technical steps may be fragmented or inconsistent. Business context is minimally addressed.	Incomplete ML pipeline with key steps missing (e.g., no evaluation or poorly applied preprocessing). Algorithm use is simplistic or generic. Demonstrates limited understanding of how ML relates to the business case.	Weak and disjointed attempt to apply machine learning. Missing coherence between method, data, and business need. Lacks clear workflow structure, justification, or meaningful outcomes..	No valid ML process presented. Critical components are missing or incorrect (e.g., no preprocessing, wrong algorithm type). Shows no alignment with dataset or business objective. Little to no technical understanding demonstrated.

DESCRIPTOR	A (100.0-93.0)	A- (92.0-88.0)	B+ (87.0-84.0)	B (83.0-80.0)	B- (79.0-77.0)	C+ (76.0-74.0)	C (73.0-72.0)	C- (71.0-70.0)	F (69.0-0.0)
CLO4 Formulate solutions to overcome challenges associated with new machine learning problem settings in a business environment	Creatively formulates solutions to emerging or complex machine learning challenges in a business context. Demonstrates advanced ability to adapt ML models to real-world constraints .. Solutions reflect deep understanding of ML limitations and business impacts, with clear pathways for integration, performance monitoring, and improvement.	Proposes innovative and context-aware solutions to machine learning challenges. Shows strong ability to identify business constraints and adjust models accordingly. Demonstrates very good understanding of limitations and proposes practical steps for model integration and evaluation, though with minor gaps in detail or originality.	Identifies key ML implementation issues and suggests feasible solutions relevant to the business case. Reasoning is sound, and adaptation to real-world constraints is evident but lacks deeper critique or broader innovation. Solutions are technically and contextually appropriate, with room for refinement.	Addresses common ML challenges with appropriate solutions that are generally well-linked to the business context. Some consideration of limitations and constraints is included, but application remains basic or moderately generic.	Suggests plausible solutions to basic ML issues. Some understanding of business challenges is present, but technical and contextual reasoning are limited or somewhat vague.	Identifies generic ML challenges but responses are oversimplified or only partially relevant. Weak connection to real business needs; limited depth in problem framing or proposed adaptation.	Addresses ML problems superficially with standard responses. Very little evidence of tailoring to real-world constraints or business-specific needs. Solutions are basic and lack insight.	Provides underdeveloped or unclear responses to ML challenges. Fails to connect technical aspects with business reality. Solutions are poorly justified or disconnected from context.	No meaningful attempt to formulate solutions to machine learning challenges. Responses show little understanding of business or technical constraints. Misinterprets or omits key aspects of the problem altogether.
Referencing %	Provides excellent referencing of literature sources, including comprehensive independent literature searches.	Provides excellent referencing of literature sources, including comprehensive independent literature searches.	Utilises and cites research sources provided appropriately. Evidence of effective independent research of the literature.	Utilises and cites research sources provided with some errors. Evidence of adequate independent research of the literature.	Adequate use and citation of research sources provided. Some evidence of own research of the literature.	Poor use and citation of research sources provided. Inadequate use of own research of the literature.	Limited use of research sources provided. Little evidence of research of the literature having been undertaken.	Very limited use of research sources provided. Very little evidence of research of the literature having been undertaken.	Fails to use or cite research sources provided. Absence of own research of the literature.
Writing style and presentation %	Exceptionally well structured, written and presented work. Reflects outstanding clarity and conciseness	Very well structured, written and presented work. Reflects very good clarity and conciseness	Generally well structured, written and presented work. Reflects good overall clarity and conciseness	Satisfactory structured, written and presented work. Reflects satisfactory clarity and conciseness	Adequate structured, written and presented work. Reflects adequate clarity and conciseness	Relatively poorly structured, written and presented work. Reflects or level of clarity and conciseness	Inadequately structured, written and presented work. Reflects inadequate clarity and conciseness	Very poorly structured, written and presented work. Reflects or level of clarity and conciseness	Highly inadequately structured, written and presented work. Lacks clarity and conciseness



Assessment Marking Scheme:

Grading Scale

LETTER CODE	PERCENTAGE
A	93.0 – 100.0
A-	88.0 – 92.0
B+	84.0 – 87.0
B	80.0 – 83.0
B-	77.0 – 79.0
C+	74.0 – 76.0
C	72.0 – 73.0
C-	70.0 – 71.0
F	Less than 69.0

7. Recommended Reading (Guide)

Lecture slides, reading materials, and text book.